

20

Question 1 (12 Marks)

Marks

(a) Evaluate  $\sqrt{\frac{53.9 \times 17.8}{19.6 + 4.97}}$  correct to 2 decimal places.

1

(b) Solve the following equations:

(i)  $8^x = \frac{1}{4}$

2

(ii)  $\frac{6}{x-1} - \frac{3}{x} = 2$

3

(c) Find a primitive of  $8 + \frac{3}{e^x}$

2

(d) Simplify:  $\frac{2\sqrt{3}}{\sqrt{3}-1}$

2

(e) A store makes a profit of 40% on the cost of all its sales. Find

(i) the selling price if the cost price is \$84

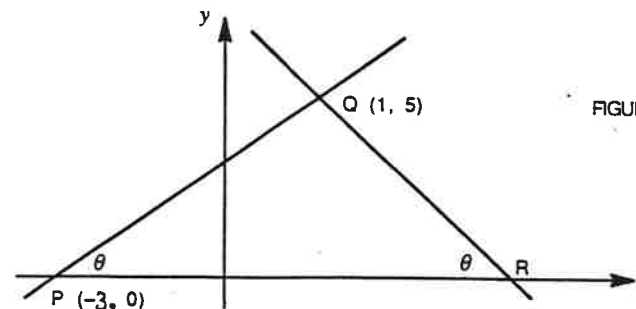
1

(ii) the cost price if the selling price is \$84.

1

Question 2 (12 Marks)

Marks



In the diagram P and Q have coordinates  $(-3, 0)$  and  $(1, 5)$  respectively, and  $\angle QPR = \angle QRP = \theta$ .

Copy this diagram onto your ANSWER SHEET.

(a) Find the coordinates of the midpoint of PQ.

1

(b) Show that the gradient of PQ is  $\frac{5}{4}$ .

1

(c) Show that the equation of PQ is  $5x - 4y + 15 = 0$ .

2

(d) Show that the gradient of QR is  $-\frac{5}{4}$ .

1

(e) Show that the equation of QR is  $5x + 4y - 25 = 0$ .

2

(f) Find the coordinates of R.

1

(g) Hence find the perpendicular distance from R to PQ.

2

(h) On your diagram, shade in the region satisfying both the inequalities:  $5x - 4y + 15 \leq 0$  and  $5x + 4y - 25 \geq 0$  simultaneously.

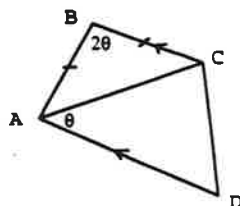
2

Question 3 (12 Marks)

Start a new page.

Marks

(a)

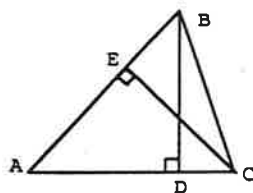


In the diagram, ABCD is a quadrilateral with  $AB = BC$ ,  $\angle ABC = 2\theta$  and  $\angle CAD = \theta$ .

Copy this diagram onto your ANSWER SHEET and find the value of  $\theta$ , giving reasons.

3

(b)



In the diagram  $BD \perp AC$  and  $CE \perp AB$ .

(i) Copy this diagram onto your ANSWER SHEET and prove that  $\triangle ECA$  is similar to  $\triangle DBA$ .

2

(ii) If  $AB = 10\text{cm}$ ,  $BD = 7\text{cm}$  and  $AC = 16\text{cm}$ , find the length of  $CE$ .

2

Question 3 continues over page

Question 3 continued

(c)

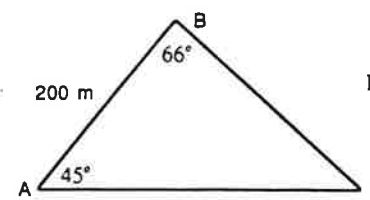


FIGURE NOT TO SCALE

Use the Sine Rule to calculate the length of the side BC to the nearest metre.

2

(d) Solve the equation:  $2\sin\theta - 1 = 0$  for  $0^\circ \leq \theta \leq 360^\circ$ .

3

Question 4 (12 Marks) Start a new page.

Marks

(a) Integrate with respect to  $x$ :

(i)  $\int (2x-1)^6 dx$  1

(ii)  $\int \sin 2x dx$  1

(b) Integrate and leave answer in exact form:  $\int_0^2 e^{2x} dx$  2

(c) Find the exact area of the region bounded by the curve  $y = \frac{1}{x+4}$ , the  $x$ -axis and the lines  $x = 0$  and  $x = 4$ , giving your answer in the simplest form.

3

(d) (i) Differentiate  $y = x^2 e^{-3x}$  with respect to  $x$ . 2

(ii) Find the equation of the tangent to the curve  $y = x^2 e^{-3x}$  at  $x = \frac{1}{3}$ .  
Leave your answer in exact form.

3

Question 5 (12 Marks)

Start a new page.

Marks

(a) Given that  $\alpha$  and  $\beta$  are the roots of the quadratic function  $2x^2 - 3x - 5 = 0$  find

(i)  $\alpha + \beta$  1

(ii)  $\alpha\beta$  1

(iii)  $(\alpha + 1)(\beta + 1)$  1

(b) Prove that the line  $y = 4x - 4$  is a tangent to  $y = x^2$ . 2

(c) The mass  $M$  kg of a radioactive substance present after  $t$  years is given by  $M = 10e^{-kt}$ , where  $k$  is a positive constant. After 100 years the mass has reduced to 5 kg.

(i) What was the initial mass? 1

(ii) Find the value of  $k$  (leave answer in exact form). 2

(iii) What amount of the radioactive substance would remain after a period of 1000 years? Give answer correct to 4 significant figures.

2

(iv) How long would it take for the initial mass to reduce to 8 kg? Give answer in terms of years, correct to 2 decimal places.

2

Question 6 (12 Marks) Start a new page.

Marks

(a) An office worker is employed at an initial salary of \$20 600 p.a. After each year this salary is increased by \$800.

(i) What is the salary for the seventh year of service? 2

(ii) What is the worker's total salary for the first seven years? 2

(b) Find the number "n" which when added to each of 2, 5 and 9 will give a set of three numbers in geometric progression.

2

(c) Show that the following represents an arithmetic series and hence evaluate

$$\sum_{k=1}^9 (4k - 1)$$

2

(d) A ball is dropped from a height of 20cm and continues to bounce  $\frac{3}{4}$  of the height of the preceding bounce until it comes to rest. What is the total distance travelled by the ball?

4

Question 7 (12Marks) Start a new page.

Marks

(a) The minute hand of a clock is 20cm long.

(i) Show that the arc length along which the tip of the hand travels in 16 minutes is  $\frac{32}{3}\pi$  radians. 1

(ii) Calculate the shortest distance between the initial and final positions of the tip of the hand. Give answer correct to 2 decimal places.

2

(b)

|      |   |     |     |     |     |
|------|---|-----|-----|-----|-----|
| x    | 1 | 2   | 3   | 4   | 5   |
| f(x) | 0 | 0.3 | 0.5 | 0.6 | 0.7 |

Given the above table, where  $f(x) = \log_{10} x$ , use Simpson's Rule with the 5 function values to find the approximate area under the curve between  $x = 1$  and  $x = 5$  (correct to 1 decimal place).

2

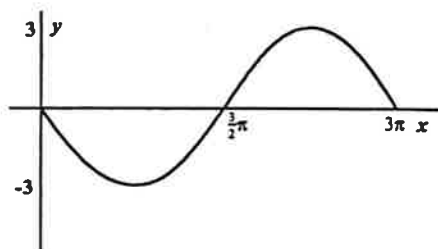
(c) A sheep, grazing in a paddock, is tethered to a stake by a rope 20m long. If the stake is 10m from a fence, find the area over which the sheep can graze. Give answer correct to 2 decimal places.

4

Question 7 continues over page

Question 7 continued

(d)



The above diagram represents a possible sine or cosine curve.

- |                                                     |   |
|-----------------------------------------------------|---|
| (i) Give the amplitude.                             | 1 |
| (ii) Give the period.                               | 1 |
| (iii) Write down a possible equation for the curve. | 1 |

Question 8 (12 Marks) Start a new page.

Marks

- |       |                                                                                                                                                                                                                                                                                                        |   |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| (a)   | Given the function $y = x^3 + 3x^2 - 24x + 1$                                                                                                                                                                                                                                                          |   |
| (i)   | find any maximum or minimum turning points                                                                                                                                                                                                                                                             | 4 |
| (ii)  | find any points of inflexion                                                                                                                                                                                                                                                                           | 2 |
| (iii) | determine where the graph cuts the y-axis                                                                                                                                                                                                                                                              | 1 |
| (iv)  | hence sketch the curve indicating its essential features.<br>Note: x intercepts not required.                                                                                                                                                                                                          | 2 |
| (b)   | Jane makes equal annual contributions of \$P into a retirement fund. This retirement fund pays compound interest of 8% per annum compounded annually. The first contribution was made on the 1 <sup>st</sup> January 2001 and the last contribution is to be made on the 1 <sup>st</sup> January 2025. |   |
| (i)   | How much does the first contribution amount to at the end of n years after interest is paid?                                                                                                                                                                                                           | 1 |
| (ii)  | If Jane wants to retire with a lump sum of \$200 000 on the 31 <sup>st</sup> December 2025 after interest is paid for the year, find the amount of each equal annual contribution \$P.                                                                                                                 | 2 |

Question 9 (12 Marks) Start a new page.

Marks

(a) Solve for  $x$ :  $9^x - 10.3^x + 9 = 0$

2

(b) A particle initially at rest at the origin moves in a straight line with velocity  $v$  metres per second, such that  $v = 3t(4-t)$  where  $t$  is the time elapsed in seconds. Find:

(i) the acceleration of the particle at the end of 1 second

1

(ii) an expression for the displacement  $x$  of the particle in terms of  $t$

2

(iii) the particle's displacement when it is next at rest

2

(iv) the velocity of the particle when it returns to the origin and state the direction in which the particle is travelling then?

2

(v) the time taken for the particle to reach its greatest velocity

1

(vi) the distance travelled by the particle in the first 5 seconds

2

Question 10 (12 Marks)

Start a new page.

Marks

(a)

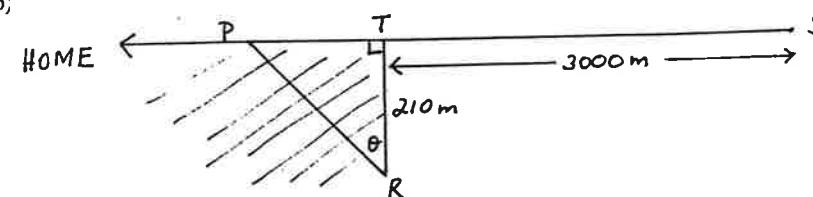
(i) Sketch the curve  $y = e^{-2x}$  and shade the region bounded by this curve, the  $x$ -axis, the  $y$ -axis and the line  $x = 2$ .

1

(ii) Find the volume generated when this area is rotated about the  $x$ -axis (give your answer in terms of  $\pi$ ).

3

(b)



The diagram above shows a restaurant  $R$  which is 210 m along a road to a point  $T$  on the main street. There is a road  $TR$  and a grassed area to walk across at an angle  $\theta$  from the restaurant to this street. Samantha must catch the last bus home after having dinner at this restaurant. The bus travels from  $S$  to  $T$  to  $P$  and beyond to home. The bus will leave the bus stop  $S$  at 10 pm sharp, travelling at a speed of  $18 \text{ m s}^{-1}$ . The distance from  $S$  to  $T$  is 3000 m. Samantha decides to walk across the grass at angle  $\theta$  and reaches the street at the point  $P$  such that  $\angle PRT = \theta$ . She walks at a speed of  $3 \text{ m s}^{-1}$ .

(i) Find  $PR$  and  $PS$  in terms of  $\theta$ .

2

(ii) Find two expressions in terms of  $\theta$ , one expression for the time taken, in seconds, for the bus to travel from  $S$  to  $P$  and the other expression for the time taken, in seconds, by Samantha to walk from  $R$  to  $P$ .

2

(iii) What is the latest that Samantha can leave the restaurant in order to catch this bus? Do not test for maximum or minimum, as we are told that there is a maximum only.

4

End of paper

## 2 UNIT TRIAL Solutions

Question 1 (a) 6.25 (to 2dp) 1

$$(b) (i) \quad 8^x = \frac{1}{4}$$

$$2^{3x} = 2^{-2}$$

$$3x = -2$$

$$x = -\frac{2}{3}$$

$$(ii) \quad \frac{6}{x-1} - \frac{3}{x} = 2$$

$$6x - 3(x-1) = 2x(x-1)$$

$$6x - 3x + 3 = 2x^2 - 2x$$

$$2x^2 - 5x - 3 = 0$$

$$(2x+1)(x-3) = 0$$

$$x = -\frac{1}{2} \text{ or } 3$$

$$(c) \quad \frac{2\sqrt{3}}{\sqrt{3}-1} = \frac{2\sqrt{3}}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} \quad \frac{1}{2}$$

$$= \frac{2\sqrt{3}(\sqrt{3}+1)}{3-1} \quad \frac{1}{2}$$

$$= \frac{2\sqrt{3}(\sqrt{3}+1)}{2} \quad \frac{1}{2}$$

$$= \sqrt{3}(\sqrt{3}+1)$$

$$= 3 + \sqrt{3} \quad \frac{1}{2}$$

$$(i) \quad S.P = 1.4 \times \$84$$

$$= \$117.60$$

$$(ii) \quad 140\% = \$84$$

$$1\% = \frac{84}{140}$$

$$100\% = \frac{84}{140} \times 100$$

$$= \$60$$

Q.2 (contd)

(d)  $m(QR) = ?$

$$\tan \theta = \frac{5}{4}$$

$$\therefore \tan(180^\circ - \theta) = -\tan \theta = -\frac{5}{4}$$

(e) Eqr. of QR:  $y - y_1 = m(x - x_1)$   
 $y - 5 = -\frac{5}{4}(x - 1)$   
 $4y - 20 = -5x + 5$   
 $5x + 4y - 25 = 0$

(f) R lies on QR  $\rightarrow 5x + 4y - 25 = 0$   
 when  $y = 0$ ,  $x = ?$

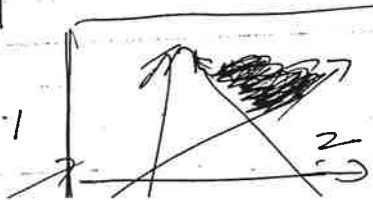
$$5x - 25 = 0$$

$$5x = 25$$

$$x = 5$$

$$\therefore R = (5, 0)$$

(g)  $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$   
 $= \frac{|5 \times 5 + (-4) \times 0 + 15|}{\sqrt{25 + 16}}$   
 $= \frac{|40|}{\sqrt{41}} = \frac{40}{\sqrt{41}}$  units



Question 3

(a)  $\angle BAC = \angle ABC = \frac{180^\circ - 2\theta}{2}$  (base  $\angle$ 's of  $\triangle ABC$  are equal and angles of  $\triangle = 180^\circ$ )

$$\angle ABC + \angle BAD = 180^\circ \text{ (Co. int } \angle\text{'s, } BC \parallel AD)$$

$$\therefore 2\theta + \frac{180^\circ - 2\theta}{2} + \theta = 180^\circ$$

$$4\theta + 180^\circ - 2\theta + 2\theta = 360^\circ$$

$$\therefore 4\theta = 180^\circ$$

$$\theta = 45^\circ$$

(b) (i) In  $\triangle ECA, DBA$

(i)  $\angle AEC = \angle ADB$  (both  $90^\circ$ , given)

(ii)  $\angle EAC = \angle BAD$  (common)

$\therefore \triangle ECA \parallel \triangle DBA$  (equiangular)

(ii)  $\frac{CE}{BD} = \frac{AC}{AB}$  (Corr. sides of similar  $\triangle$ 's are in the same ratio)

$$\therefore \frac{CE}{7} = \frac{16}{10}$$

$$\therefore CE = \frac{16 \times 7}{10} = \frac{112}{10} = 11.2$$

Qu. 3 (contd)

$$(c) \quad \frac{BC}{\sin 45^\circ} = \frac{200}{\sin 69^\circ}$$

( $\angle BCA = 180^\circ - (45^\circ + 66^\circ)$   
angle sum of  $\triangle$ )

$$\therefore BC = \frac{200 \times \sin 45^\circ}{\sin 69^\circ}$$

$$= 151.148 \dots$$

$$= 151 \text{ m (to nearest m)}$$

$$(d) \quad 2 \sin \theta - 1 = 0$$

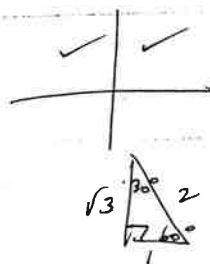
$$2 \sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

acute  $\theta = 30^\circ$

$$\therefore \theta = 30^\circ, 150^\circ$$

$$0^\circ \leq \theta \leq 360^\circ$$



Question 4

$$(a) (i) \quad \int (2x-1)^6 dx$$

$$= \frac{(2x-1)^7}{7 \times 2} + C$$

$$= \frac{(2x-1)^7}{14} + C$$

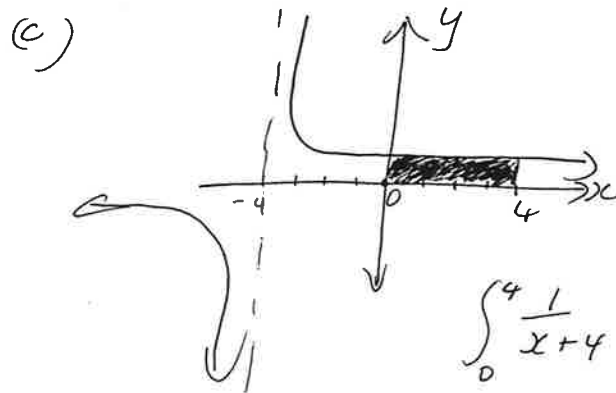
$$(ii) \quad \int \sin 2x dx$$

$$= -\frac{\cos 2x}{2} + C$$

$$(b) \quad \int_0^2 e^{2x} dx = \left[ \frac{e^{2x}}{2} \right]_0^2$$

$$= \frac{e^4}{2} - \frac{e^0}{2}$$

$$= \frac{e^4}{2} - \frac{1}{2}$$



$$\int_0^4 \frac{1}{x+4} dx = \left[ \ln(x+4) \right]_0^4$$

$$= \ln 8 - \ln 4$$

$$= 3 \ln 2 - 2 \ln 2$$

$$= \ln 2$$

Qu. 4 (contd.)

(d) (i)

$$y = x^2 e^{-3x}$$

$$y' = x^2 \cdot -3e^{-3x} + 2x \cdot e^{-3x}$$

$$y' = xe^{-3x} (2 - 3x)$$

(ii)

$$y = x^2 e^{-3x}, \quad y' = xe^{-3x} (2 - 3x)$$

$$\text{When } x = \frac{1}{3}, \quad y = \frac{1}{9} e^{-1} = \frac{1}{9e}$$

$$\text{and } y' = m = \frac{1}{3} e^{-1} (2 - 1) = \frac{1}{3e}$$

$\therefore$  Eqn of tangent to curve at  $x = \frac{1}{3}$  is:

$$y - y_1 = m(x - x_1)$$

$$y - \frac{1}{9e} = \frac{1}{3e} (x - \frac{1}{3})$$

$$9ey - 1 = 3x - 1$$

$$y = \frac{3x}{9e}$$

$$\therefore y = \frac{x}{3e}$$

Qu. 5 (contd.)

(c)

$$M = 10e^{-kt}$$

(i) When  $t = 0$ ,  $M = ?$

$$M = 10e^0$$

$$M = 10$$

$\therefore$  initial mass is 10 kg.

(ii)

When  $t = 100$ ,  $M = 5$

$$\therefore 5 = 10e^{-100k}$$

$$e^{-100k} = 0.5$$

$$\ln e^{-100k} = \ln 0.5$$

$$\therefore -100k = \ln 0.5$$

$$k = \frac{\ln 0.5}{-100}$$

(iii)

When  $t = 1000$ ,  $M = ?$

$$M = 10e^{-1000k}$$

$$= 10e^{-1000 \times \frac{\ln 0.5}{-100}}$$

$$= 0.009765625 \text{ (kg)}$$

$$= 9.766 \text{ g (4 s.f.)}$$

(iv)

When  $M = 8$ ,  $t = ?$

$$8 = 10e^{-kt}$$

$$e^{-kt} = 0.8$$

$$\ln e^{-kt} = \ln 0.8$$

$$-kt = \log 0.8$$

$$t = \frac{\log 0.8}{-k}$$

$$= 32.1940$$

## Question 5

(a)  $2x^2 - 3x - 5 = 0$

(i)  $\alpha + \beta = -\frac{b}{a} = \frac{3}{2}$

(ii)  $2\beta = \frac{c}{a} = -\frac{5}{2}$

(iii)  $(\alpha+1)(\beta+1)$

$= \alpha\beta + \alpha + \beta + 1$

$= \alpha\beta + (\alpha + \beta) + 1$

$= -\frac{5}{2} + \frac{3}{2} + 1$

$= 0$

(b)

$y = 4x - 4$ ,  $y = x^2$

Solve simultaneously

$4x - 4 = x^2$

$x^2 - 4x + 4 = 0$

$(x-2)(x-2) = 0$

$\therefore x = 2$

only one root

$\therefore y = 4x - 4$  touches  $y = x^2$

$\therefore y = 4x - 4$  is a tangent to  $y = x^2$

## Q4 b

(a) This is an A.P. with  $a = 20600$   
 $d = 800$   
 $t = ?$

(i)  $T_n = a + (n-1)d$

$T_7 = 20600 + 6 \times 800$   
 $= 25400$

(ii)  $S_n = \frac{n}{2}(a + l)$

$S_7 = \frac{7}{2}(20600 + 25400)$   
 $= \$161000$

(b)

G.P.  $\rightarrow 2+n, 5+n, 9+n$

$\therefore \frac{n+5}{n+2} = \frac{n+9}{n+5}$

$(n+5)^2 = (n+2)(n+9)$

$n^2 + 10n + 25 = n^2 + 11n + 18$   
 $7 = n$

$\therefore n = 7$

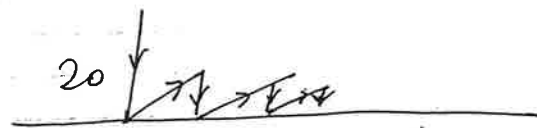
(c)

$\sum_{k=1}^9 (4k-1) = 3 + 7 + 11 + 15 + \dots$

This an AP with  $a = 3, d = 4, n = 9$

$\therefore S_9 = \frac{9}{2}(6 + 8 \times 4)$   
 $= \frac{9}{2}(38) = 171$

(d) This a G.P. with  $a = 15$ ,  $r = \frac{3}{4}$  |



$$\frac{3}{4} \times 20 = 15$$

$$\therefore a = 15, r = \frac{3}{4}$$

$$S_{\infty} = \frac{15}{\frac{1}{4}} = 60$$

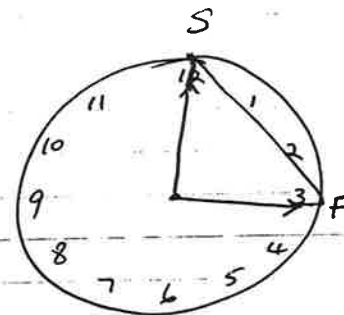
$$\therefore \text{Total distance} = 2 \times S_{\infty} + 20$$

$$= 2 \times 60 + 20$$

$$= 140 \text{ cm.}$$

Question 7

(a)(i)



$$60 \text{ min} = 360^\circ$$

$$1 \text{ min} = \frac{360}{60} = 6^\circ$$

$$\therefore 16 \text{ min} = 16 \times 6 = 96^\circ$$

$$\begin{aligned} \text{Arc length} &= \frac{96}{360} \times 2 \times \pi \times 20 \\ &= \frac{32\pi}{3} \text{ cm.} \end{aligned}$$

(ii) Shortest distance is SF.

$$\begin{aligned} SF^2 &= 20^2 + 20^2 - 2 \times 20 \times 20 \times \cos 96^\circ \\ &= 883.62277 \dots \end{aligned}$$

$$\therefore SF = 29.73 \text{ cm.}$$

7(b)  $h = \frac{5-1}{4} = \frac{4}{4} = 1$  or  $h = 1$  from  $\frac{1}{2}$  table.

$$\therefore A = \frac{1}{3} (0 + 4 \times 0.3 + 2 \times 0.5 + 4 \times 0.6 + 0.7)$$

$$\begin{aligned} A &= \frac{5.3}{3} = 1.777\dots \\ &= 1.8 \text{ unit}^2 \end{aligned}$$

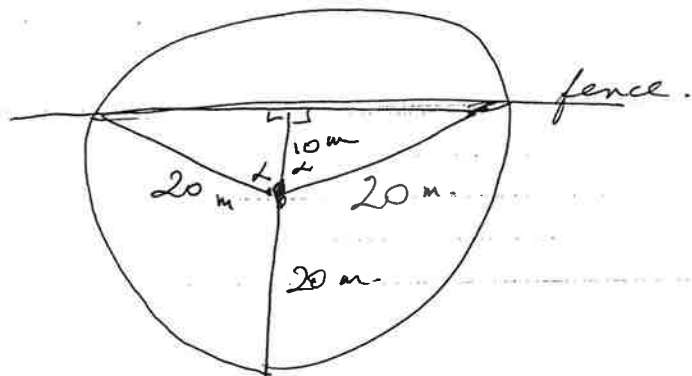
7(c) next sheet

7(d) (i) amplitude = 3

(ii) period =  $3\pi$

(iii)  $-3 \sin 2x$

(c)



$$\cos \alpha = \frac{10}{20} = \frac{1}{2}$$

$$\alpha = 60^\circ$$

$$2\alpha = 120^\circ = \frac{2\pi}{3}$$

 $\frac{1}{2}$ Let  $\theta = 2\alpha$ .

$$A \text{ of sector} = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 20^2 \times \frac{2\pi}{3} = \frac{400\pi}{3}$$

$$A \text{ of triangle} = \frac{1}{2} ab \sin \theta$$

$$= \frac{1}{2} \times 20 \times 20 \times \sin 120^\circ$$

$$= \frac{1}{2} \times 20 \times 20 \times \frac{\sqrt{3}}{2}$$

$$= 100\sqrt{3}$$

$$A \text{ of segment} = \frac{400\pi}{3} - 100\sqrt{3}$$

$$A \text{ of circle} = \pi r^2 = \pi \times 400 = 400\pi$$

$$\begin{aligned} \therefore \text{grazing area} &= 400\pi - \left( \frac{400\pi}{3} - 100\sqrt{3} \right) \\ &= 1200\pi - 400\pi - 300\sqrt{3} \\ &= 800\pi - 300\sqrt{3} \end{aligned}$$

Question 8

$$(a) (i) \quad y = x^3 + 3x^2 - 24x + 1$$

$$y' = 3x^2 + 6x - 24$$

Stat pts when  $y' = 0$ 

$$\text{i.e. } 3x^2 + 6x - 24 = 0$$

$$(\div 3) \quad x^2 + 2x - 8 = 0$$

$$(x-2)(x+4) = 0$$

$$\therefore x = 2 \text{ or } -4$$

$$y'' = 6x + 6$$

When  $x = 2$ ,  $y'' = 12 + 6$  (+ve)  $\therefore$  a min. tp.When  $x = -4$ ,  $y'' = -24 + 6$  (-ve)  $\therefore$  a max. tp.

$$\text{When } x = 2, y = -27$$

$$\text{When } x = -4, y = 81$$

 $\therefore$  a max. tp at  $(-4, 81)$  anda min. tp at  $(2, -27)$ .(ii) Pt. of inflexion when  $y'' = 0$  and there is a change of concavity.

$$\text{i.e. } 6x + 6 = 0$$

$$6x = -6$$

$$x = -1$$

$$\text{When } x = -1, y = 27$$

|       |        |      |        |
|-------|--------|------|--------|
| $x$   | $-1^-$ | $-1$ | $-1^+$ |
| $y''$ | $-$    | $0$  | $+$    |

 $\therefore$  a change in concavity $\therefore$  a pt. of inflexion at  $(-1, 27)$

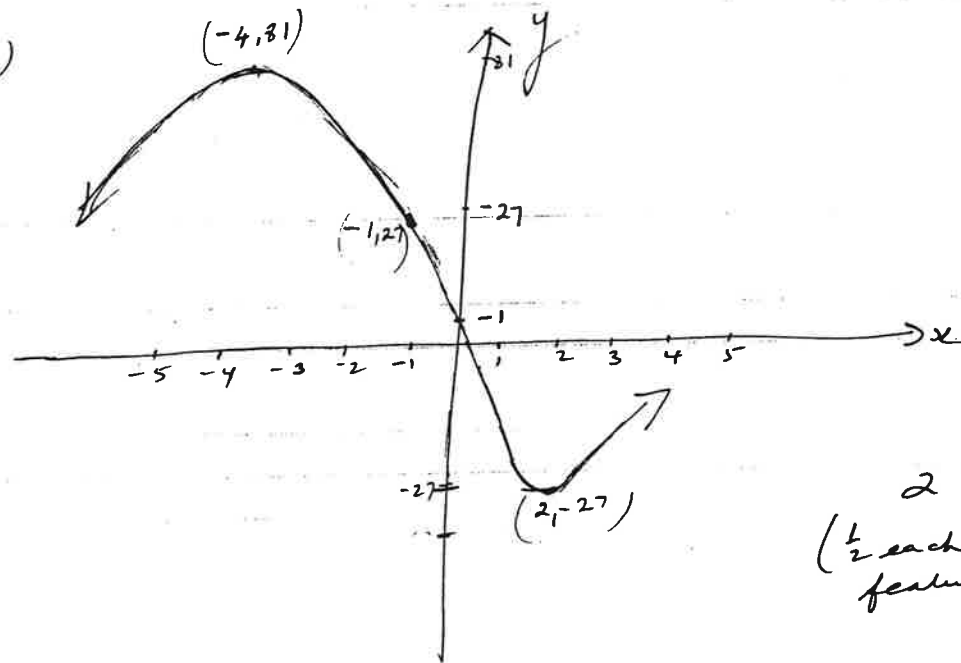
Qu. 8 contd.

(iii) When  $x = 0$ ,  $y = ?$

$$y = 1$$

$\therefore$  graph cuts  $y$ -axis at  $(0, 1)$ .

(iv)



2  
( $\frac{1}{2}$  each  
feature)

Qu. 8 contd

8 (b)  $1^{25}$

$$= \$m (1.08)^{25}$$

$2^{nd}$

$$= \$m (1.08)^{24}$$

$25^{th}$

$$= \$m (1.08)^1$$

$$200000 = m [1.08 + 1.08^2 + \dots + 1.08^{25}]$$

$$200000 = m (1.08) \frac{[1.08^{25} - 1]}{1.08 - 1}$$

$$\therefore m = \frac{200000 \times 0.8}{(1.08)(1.08^{25} - 1)}$$

$$m = \$2533.11$$

# Question 9

(a)  $9x^2 - 10 \cdot 3x + 9 = 0$

Let  $k = 3x$

$k^2 - 10k + 9 = 0$

$(k-1)(k-9) = 0$

$\therefore k = 1 \text{ or } 9$

$\therefore 3x = 1 \quad \text{or} \quad 3x = 9$

$3x = 3^0$

$\therefore x = 0$

$3x = 3^2$

$x = 2$

$\therefore x = 0 \text{ or } 2$

(b) (i)  $v = 3t(4-t)$

$v = 12t - 3t^2$

$\therefore a = 12 - 6t$

When  $t = 1$ ,  $a = 12 - 6 = 6 \text{ m/s}^2$

(ii)  $x = \int (12t - 3t^2) dt$

$= \frac{12t^2}{2} - \frac{3t^3}{3} + C$

$= 6t^2 - t^3 + C$

When  $t = 0$ ,  $x = 0$ ,  $\therefore C = 0$

$\therefore x = 6t^2 - t^3$

(iii) When  $v = 0$

$3t(4-t) = 0$

$\therefore t = 0 \text{ or } 4$

$\therefore$  it is next at rest after 4 sec.

$\therefore$  when  $t = 4$ ,  $x = 6 \cdot 4^2 - 4^3$

$= 32 \text{ m}$

(iv) When  $x = 0$ ,  $6t^2 - t^3 = 0$

$t^2(6-t) = 0$

$t = 0 \text{ or } 6$

$\therefore$  when  $t = 6$ ,  $v = 18(4-6)$   
 $= -36 \text{ m/s}$

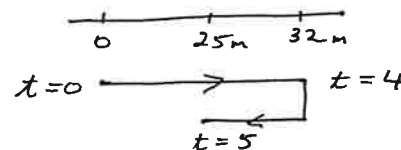
(v) Max velocity when  $a = 0$

$\therefore 12 - 6t = 0$

$t = 2$

$\therefore$  it takes 2 sec for it to reach max. velocity.

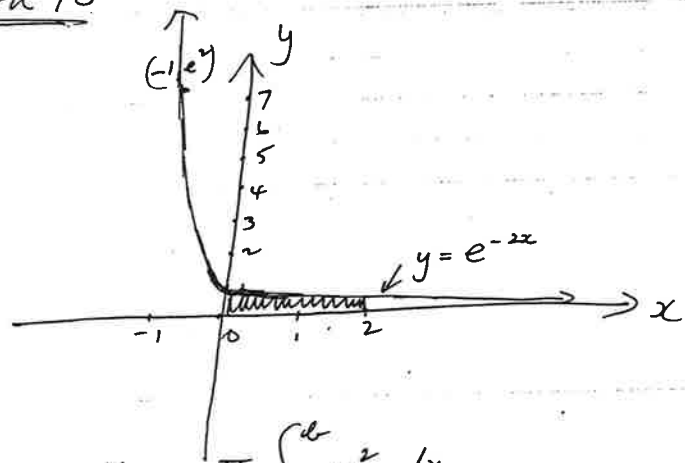
(vi) When  $t = 5$ ,  $x = 150 - 125 = 25 \text{ m}$



$\therefore$  it travels  $32 + (32 - 25) = 39 \text{ m}$  in the first 5 sec.

# Question 10

(a) (i)



2

(ii)

$$V = \pi \int_a^b y^2 dx$$

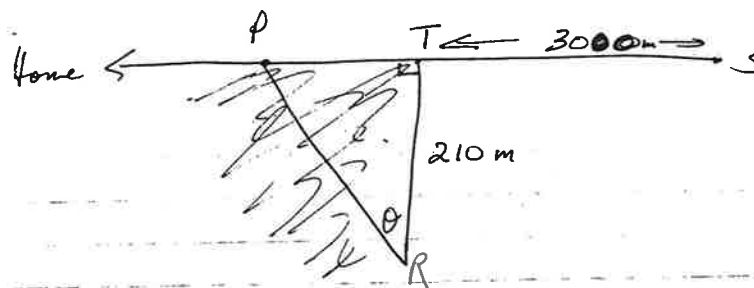
$$V = \pi \int_0^2 (e^{-2x})^2 dx$$

$$= \pi \int_0^2 e^{-4x} dx$$

$$= \pi \left[ \frac{e^{-4x}}{-4} \right]_0^2$$

$$= \pi \left( \frac{e^{-8}}{-4} + \frac{e^0}{4} \right)$$

$$= \frac{\pi}{4} \left( 1 - \frac{1}{e^8} \right) \text{ units}^3$$



$$S(\text{Samantha}) = 3 \text{ m/s}$$

$$S(\text{bus}) = 18 \text{ m/s}$$

(i)

$$\cos \theta = \frac{210}{PR}$$

$$\therefore PR = \frac{210}{\cos \theta} \text{ or } 210 \sec \theta$$

$$\tan \theta = \frac{PT}{210}$$

$$\therefore PT = 210 \tan \theta$$

$$\therefore PS = 210 \tan \theta + 3000$$

(ii)

$$t(\text{bus}) = \frac{d}{S} = \frac{210 \tan \theta + 3000}{18}$$

$$t(\text{Samantha}) = \frac{d}{S} = \frac{210}{\frac{3}{\sec \theta}} = \frac{210 \sec \theta}{3}$$

(iii)

$$T = t(\text{bus}) - t(\text{Samantha})$$

$$= \frac{210 \tan \theta + 3000}{18} - \frac{210 \sec \theta}{3}$$

$$= \frac{210 \tan \theta + 3000 - 1260 \sec \theta}{18}$$

$$\begin{aligned}\frac{dT}{d\theta} &= \frac{1}{18} (210 \sec^2 \theta - 1260 \sec \theta \tan \theta) \\ &= \frac{35 \sec^2 \theta - 210 \sec \theta \tan \theta}{3} \\ &= 35 \sec \theta \left( \frac{\sec \theta}{3} - 2 \tan \theta \right)\end{aligned}$$

When  $\frac{dT}{d\theta} = 0$ ,

$$35 \sec \theta = 0$$

$$\text{im } \sec \theta = 0 \quad (\text{no soln})$$

or  $\frac{\sec \theta}{3} - 2 \tan \theta = 0$

$$\frac{\sec \theta}{3} = 2 \tan \theta$$

$$\frac{1}{3 \cos \theta} = \frac{2 \sin \theta}{\cos \theta}$$

$$\therefore \sin \theta = \frac{1}{6}$$

$$\therefore \theta = 9.46^\circ \quad (10 \text{ nearest min})$$

$$\text{or } \theta = 0.1674480 \text{ radians}$$

$$\therefore T = \frac{210 \tan \theta + 3000 - 1260 \sec \theta}{18}$$

$$= \frac{210 \tan 0.16745 + 3000 - 1260 \sec 0.1674}{18}$$

$$= 97.64514 \text{ secs. after } 10 \text{ pm.}$$

$$= 1 \text{ min } 38 \text{ secs. after } 10 \text{ pm (using radians)}$$

or  $1 \text{ min } 39 \text{ secs after } 10 \text{ pm (using degrees)}$